

String Practice Questions (150)

1. Find the length of a string.
2. Copy one string to another.
3. Concatenate two strings.
4. Compare two strings (case-sensitive).
5. Compare two strings ignoring case.
6. Convert a string to uppercase.
7. Convert a string to lowercase.
8. Toggle the case of each character in a string.
9. Check whether a string is empty.
10. Trim leading, trailing, or extra spaces from a string.
11. Get the character at a given index.
12. Get the Unicode code point of a character at an index.
13. Get the Unicode code point before a given index.
14. Find the first occurrence of a character in a string.
15. Find the last occurrence of a character in a string.
16. Count total occurrences of a character in a string.
17. Remove the first, last, or all occurrences of a given character.
18. Replace the first, last, or all occurrences of a character.
19. Find the highest frequency character in a string.
20. Find the lowest frequency character in a string.
21. Find the first non-repeating character.
22. Find the last repeating character.
23. Print all characters that occur exactly twice.
24. Check if all characters in a string are unique (no repetition).
25. Count total number of words in a string.
26. Find the first occurrence of a word in a string.
27. Find the last occurrence of a word in a string.
28. Count occurrences of a word in a string.
29. Remove the first, last, or all occurrences of a word.
30. Replace a word with another word.
31. Remove duplicate words from a string.
32. Count the frequency of each word in a string.
33. Find the longest word in a string.
34. Find the shortest word in a string.
35. Find the first word that is a palindrome.
36. Reverse the order of words in a string.

37. Reverse each word in a string.
38. Reverse words without using `split()`.
39. Search all occurrences of a character.
40. Search all occurrences of a word.
41. Check if a string contains a substring (without using `contains()`).
42. Check if two strings are equal without using `equals()`.
43. Check if two strings are rotations of each other.
44. Check if two strings are anagrams.
45. Check whether a string starts with or ends with another string.
46. Check if a substring appears at both the start and end.
47. Check if one string is a substring of another using only concatenation.
48. Remove all vowels from a string.
49. Replace all consonants with '*'.
50. Remove all digits from a string.
51. Extract only digits from a string.
52. Remove all special characters.
53. Remove all punctuation characters.
54. Replace all duplicate characters with '\$'.
55. Reverse only vowels in a string.
56. Reverse only consonants in a string.
57. Merge two strings alternatively (char by char).
58. Rotate characters by 2 positions to the left.
59. Rotate characters by 3 positions to the right.
60. Append two strings but remove duplicate adjacent characters.
61. Count total alphabets, digits, and special characters.
62. Count vowels and consonants.
63. Count frequency of each character.
64. Count frequency of each vowel.
65. Count palindromic substrings in a string.
66. Count number of sentences in a paragraph.
67. Count how many times a substring appears in a string.
68. Count the sum of digits present in a string.
69. Count how many times 'life' appears in a string.
70. Compare the number of times 'the' and 'is' appear.
71. Print all substrings of a string.
72. Print all substrings of length n.
73. Find the longest palindromic substring.
74. Find the longest substring without repeating characters.

75. Find the longest common prefix among strings.
76. Find the longest common suffix among strings.
77. Find the longest substring that appears at both ends.
78. Find the longest mirror-image substring at both ends.
79. Divide a string into n equal parts.
80. Print list items containing all characters of a given word.
81. Generate a hash code or UUID for a given string.
82. Create a string from a character array.
83. Create a string from a byte array.
84. Print ASCII value of each character.
85. Convert string into a char array without built-in functions.
86. Print all permutations of a string without repetition.
87. Print all permutations of a string with repetition.
88. Rearrange a string so that all identical characters are at least d distance apart.
89. Remove 'b' and 'ac' from a string.
90. Remove adjacent duplicates recursively.
91. Check if two strings are interleaving of another string.
92. Check if two strings are pq-balanced.
93. Match strings with wildcard characters (*, ?).
94. Find the smallest window containing all characters of another string.
95. Find the second most frequent character in a string.
96. Find the second most frequent word in a string.
97. Check if two given strings appear at the end of each other (ignoring case).
98. Check if the first 'z' is immediately followed by another 'z'.
99. Check if a 'z' is happy (surrounded by same chars).
100. Return true if string contains 'abc' not followed by '!'.
101. Check if a string is a valid palindrome ignoring spaces and punctuation.
102. Reverse a string using recursion.
103. Check if a string contains balanced parentheses.
104. Check if a string contains balanced brackets of all types ((), {}, []).
105. Find the longest valid parentheses substring.
106. Generate all subsequences of a string.
107. Check if a string is a pangram (contains every letter).
108. Check if a string is an isogram (no repeating letters).
109. Find the lexicographically smallest substring of length k.
110. Find the lexicographically largest substring of length k.
111. Check if a string can be rearranged into a palindrome.
112. Find the minimum number of deletions to make a string palindrome.

113. Find the minimum number of insertions to make a string palindrome.
114. Check if one string is a subsequence of another.
115. Find the edit distance (Levenshtein distance) between two strings.
116. Check if a string is a valid shuffle of two other strings.
117. Check if a string contains duplicate substrings.
118. Find the longest repeated substring.
119. Find the smallest substring containing all vowels.
120. Find the longest substring containing only vowels.
121. Check if a string contains only binary digits (0/1).
122. Convert a binary string to decimal.
123. Convert a decimal number to binary string.
124. Find the longest common subsequence of two strings.
125. Find the shortest common supersequence of two strings.
126. Print all anagrams of a string.
127. Group words that are anagrams from an array of strings.
128. Check if a string follows a given pattern.
129. Find the smallest window containing all distinct characters.
130. Find the maximum occurring word in a string.
131. Check if a string is a valid email address.
132. Check if a string is a valid IP address.
133. Convert a Roman numeral string to integer.
134. Convert an integer to Roman numeral string.
135. Check if two strings differ by exactly one character.
136. Check if two strings are one edit distance apart.
137. Find the length of the longest substring with at most k distinct characters.
138. Find all palindromic partitions of a string.
139. Check if a string can be segmented into valid dictionary words.
140. Implement KMP algorithm for substring search.
141. Implement Rabin-Karp algorithm for substring search.
142. Implement Z algorithm for substring search.
143. Check if a string is valid JSON format (basic check).
144. Check if a string is valid HTML/XML tag sequence.
145. Remove HTML tags from a string.
146. Compress a string using run-length encoding.
147. Decompress a run-length encoded string.
148. Find the first recurring substring of length k.
149. Check if two strings are scrambled versions of each other.
150. Find the lexicographically next permutation of a string.

151. Find the lexicographically previous permutation of a string.